Assignment of Java Programming

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Assignment Questions

Q.1. Write a simple Java Program to print “Hello World!” to the console. And Also Explain it.

Ans.
```
public class question1 {
    public static void main(String[] args) {
        System.out.println("Hello World!");
    }
}
```

Q.2. Write a Java program to print the results of the following operations. Test Data: a. -5 + 8 * 6 b. (55+9) % 9 c. 20 + -3*5 / 8 d. 5 + 15 / 3 * 2 - 8 % 3.

Ans.
```
public class question2 {
    public static void main(String[] args) {
        System.out.println("A-> " + (-5 + 8 * 6));
        System.out.println("B-> " + ((55+9) % 9));
        System.out.println("C-> " + (20 + -3*5 / 8));
        System.out.println("D-> " + (5 + 15 / 3 * 2 - 8 %
3));
    }
}
```

Q.3. Write a Java program to print the sum (addition), multiply, subtract, divide and remainder of two numbers.

Ans.
```
public class question3 {
    public static void main(String[] args) {
        int a = 6;
```

```
        int b = 4;
        System.out.println("Addition :- " + (a+b));
        System.out.println("Subtration :- " + (a-b));
        System.out.println("Multiplication :- " + (a*b));
        System.out.println("Division :- " + (a/b));
        System.out.println("Remainder :- " + (a%b));
    }
}
```

Q.4. Write a program for society system whether user enter the flat number and got a flat owner details to the console. (Note: Make this by nested if-else & Switch case both).

Ans.
```
import java.util.Scanner;
public class question4 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Flat Number (e.g., 101, 
102, 201, 202): ");
        int flatNumber = sc.nextInt();
        System.out.println("-----Using Nested-----");
        if (flatNumber==101) {
            System.out.println("Owner: Mr. Sharma");
        }
        else if(flatNumber == 102) {
            System.out.println("Owner: Mr. Verma");
        }
        else if(flatNumber == 201) {
            System.out.println("Owner: Mr. Pandya");
        }
```

```java
            else if(flatNumber == 202) {
                System.out.println("Owner: Mr. Gupta");
        }
        else {
            System.out.println("Invalid flat number!");
        }
        System.out.println("-----Using Switch Case-----");
        switch(flatNumber){
            case 101:
            System.out.println("Owner: Mr. Sharma");
            break;
            case 102:
            System.out.println("Owner: Mr. Verma");
            break;
            case 201:
            System.out.println("Owner: Mr. Pandya");
            break;
            case 202:
            System.out.println("Owner: Mr. Gupta");
            break;
            default :
            System.out.println("Invalid flat number!");
          }
        sc.close();
    }
}
```

Q.5. Write a program for making the grading system whether user enter the percentage then got a grade to the console.

Ans.
```
import java.util.Scanner;
public class question5 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter your percentage: ");
        double percentage = sc.nextDouble();
        if (percentage >= 90 && percentage <= 100) {
            System.out.println("Grade: A+");
        }
        else if (percentage >= 80 && percentage < 90) {
            System.out.println("Grade: A");
        }
        else if (percentage >= 70 && percentage < 80) {
            System.out.println("Grade: B");
        }
        else if (percentage >= 60 && percentage < 70) {
            System.out.println("Grade: C");
        }
        else if (percentage >= 50 && percentage < 60) {
            System.out.println("Grade: D");
        }
        else if (percentage >= 0 && percentage < 50) {
                System.out.println("Grade: F (Fail)");
        }
        else {
            System.out.println("Invalid percentage
entered!");
        }
        sc.close();
```

```
    }
}
```

Q.6. Write a program for making month system whether user input the month number and got a month name.

Ans.
```
import java.util.Scanner;
public class question6 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter month number (1 to 12): ");
        int month = sc.nextInt();
        switch (month) {
            case 1:
                System.out.println("Month: January");
                break;
            case 2:
                System.out.println("Month: February");
                break;
            case 3:
                System.out.println("Month: March");
                break;
            case 4:
                System.out.println("Month: April");
                break;
            case 5:
                 System.out.println("Month: May");
                break;
            case 6:
                 System.out.println("Month: June");
```

```java
                    break;
                case 7:
                    System.out.println("Month: July");
                    break;
                case 8:
                    System.out.println("Month: August");
                    break;
                case 9:
                    System.out.println("Month: September");
                    break;
                case 10:
                    System.out.println("Month: October");
                    break;
                case 11:
                    System.out.println("Month: November");
                    break;
                case 12:
                    System.out.println("Month: December");
                    break;
                default:
                    System.out.println("Invalid month number!
Please enter between 1 and 12.");
        }
        sc.close();
    }
}
```

Q.7. Write a program for clearing three steps security check whether user input 1st PIN number then asked 2nd PIN and finally asked 3rd PIN then got a login into system.

Ans.
```
import java.util.Scanner;
public class question7 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        // Define the correct PINs
        int pin1 = 1234;
        int pin2 = 5678;
        int pin3 = 9012;
        System.out.print("Enter 1st PIN: ");
        int input1 = sc.nextInt();
        if (input1 == pin1) {
            System.out.print("Enter 2nd PIN: ");
            int input2 = sc.nextInt();
            if (input2 == pin2) {
                System.out.print("Enter 3rd PIN: ");
                int input3 = sc.nextInt();
                if (input3 == pin3) {
                    System.out.println("Login Successful.
Access Granted!");
                }
                else {
                    System.out.println("Incorrect 3rd PIN.
Access Denied.");
                }

            }
            else {
```

```
                     System.out.println("Incorrect 2nd PIN.
Access Denied.");
              }
        }
        else {
            System.out.println("Incorrect 1st PIN. Access
Denied.");
        }
        sc.close();
    }
}
```

Q.8. Write a program where you asked the user which number of table is required then print the table is format of '2 X 1 = 2' using all types of loop (for, while and do-while loop).

Ans.
```
import java.util.Scanner;
public class question8 {
    public static void main(String[] args) {
        int a = 1;
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the number for table : ");
        int num = sc.nextInt();
        System.out.println("\n-----Using for loop-----");
        for(int i=num;i<=num*10;i=i+num){
            System.out.println(num +" X "+(a++) +" = "+ i
+ " ");
        }
        System.out.println("\n-----Using while loop-----");
          int i = num;
```

```
        int b = 1;
        while (i<=num*10) {
                System.out.println(num +" X "+(b++) +" = "+
i + " ");
            i = i+num;
        }
        System.out.println("\n-----Using do-while loop----
-");
        int j = num;
        int c = 1;
        do{
            System.out.println(num +" X "+(c++) +" = "+ j
+ " ");
            j = j+num;
        } while(j<=num*10);
        sc.close();
    }
}
```

Q.9. Write a program where you print only even numbers from 1 to 100.

Ans.
```
public class question9 {
    public static void main(String[] args) {
        for(int i=1;i<=100;i++){
            if(i%2==0){
                System.out.print(i + " ");
            }
        }
    }
```

```
}
```

Q.10. Write a program whether the input number is prime or not.

Ans.
```
import java.util.Scanner;
public class question10 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number : ");
        int number = sc.nextInt();
        boolean isPrime = true;
        if (number <= 1) {
            isPrime = false;
        }
        else {
            for (int i = 2; i <= Math.sqrt(number); i++) {
                if (number % i == 0) {
                    isPrime = false;
                    break;
                }
            }
        }
         if (isPrime) {
            System.out.println(number + " is a Prime
number.");
        }
        else {
            System.out.println(number + " is not a Prime
number.");
```

```
        }
          sc.close();
    }
}
```

Q.11. Write a program for make those patterns given below:-

1) #
# #
# # #
# # # #
# # # # #

```
Ans. public class question11a {
    public static void main(String[] args) {
        int n = 5;
        for(int i=1;i<=n;i++){
            for(int j=1;j<=i;j++){
                System.out.print("#" + " ");
            }
            System.out.println();
        }
    }
}
```

2) # # # # #
# # # #
# # #
# #
#

```
Ans. public class question11b {
```

```
    public static void main(String[] args) {
        int n = 5;
          for(int i=1;i<=n;i++){
            for(int j=i;j<=n;j++){
                System.out.print("#" + " ");
            }
            System.out.println();
        }
    }
}
```

3) 1
   1 2
   1 2 3
   1 2 3 4
   1 2 3 4 5

Ans.
```
public class question11c {
    public static void main(String[] args) {
        int n = 5;
        for(int i=1;i<=n;i++){
            for(int j=1;j<=i;j++){
                System.out.print(j + " ");
            }
            System.out.println();
        }
    }
}
```

4) 5 5 5 5 5
   4 4 4 4

3 3 3
2 2
1

Ans.
```java
public class question11d {
    public static void main(String[] args) {
        int n = 5;
        for(int i=n;i>=1;i--){
            for(int j=1;j<=i;j++){
                System.out.print(i + " ");
            }
            System.out.println();
        }
    }
}
```

5) #
# #
# # #
# # # #
# # # # #
# # # # #
# # # #
# # #
# #
#

Ans.
```java
public class question11e {
    public static void main(String[] args) {
        int n = 5;
        for(int i=1;i<=n;i++){
            for(int j=1;j<=i;j++){
                System.out.print("#" + " ");
```

```
            }
            System.out.println();
          }
        for(int i=1;i<=n;i++){
            for(int j=i;j<=n;j++){
                System.out.print("#" + " ");
            }
            System.out.println();
        }
    }
}
```

6) # # # # #
   # # # #
   # # #
   # #
   #
   # #
   # # #
   # # # #
   # # # # #

Ans.
```
public class question11f {
    public static void main(String[] args) {
        int n = 5;
        for(int i=1;i<=n;i++){
            for(int j=i;j<=n;j++){
                System.out.print("#" + " ");
            }
            System.out.println();
        }
```

```
        for(int i=2;i<=n;i++){
            for(int j=1;j<=i;j++){
                   System.out.print("#" + " ");
            }
            System.out.println();
        }
    }
}
```

Q.12. Write a Java program to sum values of an array.

Ans.
```
public class question12 {
    public static void main(String[] args) {
        int[] arr = {1,2,3,4,5};
        int sum = 0;
        for(int i=0;i<=arr.length;i++){
            sum = sum + i;
        }
        System.out.println("Sum :- " + sum);
    }
}
```

Q.13. Write a Java program to calculate the average value of array elements.

Ans.
```
public class question13 {
    public static void main(String[] args) {
        int[] arr = {1,2,3,4,5};
        int sum = 0;
        for(int i=0;i<=arr.length;i++){
```

```
            sum = sum + i;
        }
          int average = sum/arr.length;
        System.out.println("Average :- " + average);
    }
}
```

Q.14. Write a program for taking two string values from user and compare it.

Ans.
```
import java.util.Scanner;
public class question14 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the first string : ");
        String s = sc.nextLine();
        System.out.print("Enter the second string : ");
        String s1 = sc.nextLine();
        if (s.equals(s1)) {
            System.out.println("Equal");
        }
        else {
            System.out.print("Not Equal");
        }
        sc.close();
    }
}
```

Q.15. Write a program for taking first name and last name from user and show full name of the user.

Ans.
```java
import java.util.Scanner;
public class question15 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the first name : ");
        String fname = sc.nextLine();
        System.out.print("Enter the last name : ");
        String lname = sc.nextLine();
        System.out.println(fname +" "+ lname);
        sc.close();
    }
}
```

Q.16. Write a Java recursive method to calculate the factorial of a given positive integer.

Ans.
```java
public class question16 {
    public static int factorial(int n){
        if(n==0) return 1;
        return n*factorial(n-1);
    }
    public static void main(String[] args) {
        int a = 5;
        System.out.println("Factorial :- " + factorial(a));
    }
}
```

Q.17. Create a program to read data from a text file and write data to a new text file.

Ans.
```java
import java.io.*;
public class question17 {
    public static void main(String[] args) {
        String inputFile = "input.txt";
        String outputFile = "output.txt";
        try (
            BufferedReader reader = new BufferedReader(new
FileReader(inputFile));
            BufferedWriter writer = new BufferedWriter(new
FileWriter(outputFile))
        ) {
            String line;
            while ((line = reader.readLine()) != null) {
                writer.write(line);
                writer.newLine();
            }
            System.out.println("File copied successfully
from " + inputFile + " to " + outputFile);
        } catch (IOException e) {
            System.out.println("An error occurred: " +
e.getMessage());
        }
    }
}
```

Q.18. Write a Java program to create a class called "Cat" with instance variables name and age. Implement a default constructor that initializes the name to "Unknown" and the age to 0. Print the values of the variables.

```
Ans. class Cat{
     String name;
    int age;
    public Cat(){
        name = "Unknown";
        age = 0;
    }
    public void displayInfo(){
        System.out.println("Cat's Name: " + name);
        System.out.println("Cat's Age: " + age);
    }
}
public class question18 {
    public static void main(String[] args) {
        Cat myCat = new Cat();
        myCat.displayInfo();
    }
}
```

Q.19. Write a Java program to create an outer class called Coputer with an inner class Processor. The Processor class should have a method "displayDetails()" that prints the details of the processor (e.g., brand and speed). Create an instance of Processor from the Computer class and call the "displayDetails()" method.

```
Ans. class Computer{
    class Processor{
        String brand = "Intel";
        double speed = 3.5;
        public void displayInfo() {
```

```
            System.out.println("Processor Brand: " +
brand);
            System.out.println("Processor Speed: " + speed
+ " GHz");
        }
    }
}
public class question19 {
    public static void main(String[] args) {
        Computer c = new Computer();
        Computer.Processor p = c.new Processor();
        p.displayInfo();
    }
}
```

Q.20. Write a Java program to create a class called Animal with a method called makeSound(). Create a subclass called Cat that overrides the makeSound() method to bark.

```
Ans. class Animal{
    public void makeSound() {
        System.out.println("Animal makes a sound");
    }
}
class Cat extends Animal{
    @Override
    public void makeSound() {
        System.out.println("Cat barks");
    }
}
```

```java
public class question20 {
    public static void main(String[] args) {
        Cat c = new Cat();
        c.makeSound();
    }
}
```

Q.21. Write a Java program to create an abstract class Animal with an abstract method called sound(). Create subclasses Lion and Tiger that extend the Animal class and implement the sound() method to make a specific sound for each animal.

```java
Ans. abstract class Animal{
    public abstract void sound();
}
class Lion extends Animal{
    @Override
    public void sound() {
        System.out.println("Lion barks");
    }
}
class Tiger extends Animal{
    @Override
    public void sound() {
        System.out.println("Tiger barks");
    }
}
public class question21 {
    public static void main(String[] args) {
        Animal lion = new Lion();
```

```
        Animal tiger = new Tiger();
        lion.sound();
        tiger.sound();
    }
}
```

Q.22. Write a Java program to create an interface Shape with the getArea() method. Create three classes Rectangle, Circle, and Triangle that implement the Shape interface. Implement the getArea() method for each of the three classes.

Ans.
```
interface Shape{
    double getArea();
}
class Rectangle implements Shape{
    double length, width;
    public Rectangle(double length,double width){
        this.length = length;
        this.width = width;
    }
    public double getArea(){
        return length * width;
    }
}
class Circle implements Shape{
    double radius;
    public Circle(double radius){
        this.radius = radius;
    }
    public double getArea(){
```

```java
        return Math.PI*radius*radius;
    }
}
class Triangle implements Shape{
    double base,height;
    public Triangle(double base, double height){
        this.base = base;
        this.height = height;
    }
    public double getArea(){
        return 0.5*base*height;
    }
}
public class question22 {
    public static void main(String[] args) {
        Rectangle r = new Rectangle(5, 3);
        Circle c = new Circle(4);
        Triangle t = new Triangle(6, 2);
        System.out.println("Area of Rectangle: " +
r.getArea());
        System.out.println("Area of Circle: " +
c.getArea());
        System.out.println("Area of Triangle: " +
t.getArea());
    }
}
```

Q.23. Write a Java program to create a class called Person with private instance variables name, age. and country. Provide public getter and setter methods to access and modify these variables.

```
Ans. class Person{
    private String name;
    private int age;
    private String country;
    public String getName() {
        return name;
    }
    public void setName(String name) {
        this.name = name;
    }
    public int getAge() {
        return age;
    }
    public void setAge(int age) {
        this.age = age;
    }
    public String getCountry() {
        return country;
    }
    public void setCountry(String country) {
        this.country = country;
    }
}
public class question23 {
    public static void main(String[] args) {
        Person p = new Person();
        p.setName("Joe Root");
        p.setAge(20);
        p.setCountry("Chile");
```

```
        System.out.println("Name:- " + p.getName());
        System.out.println("Age:- " + p.getAge());
        System.out.println("Country:- " +
p.getCountry());
    }
}
```

Q.24. Write a Java program to create a base class Animal (Animal Family) with a method called Sound(). Create two subclasses Bird and Cat. Override the Sound() method in each subclass to make a specific sound for each animal.

```
Ans. class Animal{
    public void sound(){
        System.out.println("Animal makes a sound");
    }
}
class Bird extends Animal{
    @Override
    public void sound(){
        System.out.println("Bird makes a sound");
    }
}
class Cat extends Animal{
    @Override
    public void sound(){
        System.out.println("Cat makes a sound");
    }
}
public class question24 {
```

```java
    public static void main(String[] args) {
        Animal a = new Animal();
        Animal b = new Bird();
        Animal c = new Cat();
        a.sound();
        b.sound();
        c.sound();
    }
}
```

Q.25. Write a Java program to create a method that reads a file and throws an exception if the file is not found.

Ans.
```java
import java.io.*;
import java.util.*;
public class question25 {
     public static void main readFile(String fileName) throws
FileNotFoundException {
        File file = new File(fileName);
        Scanner sc = new Scanner(new FileReader(file));
        System.out.println("File contents:");
        while (sc.hasNextLine()) {
            System.out.println(sc.nextLine());
        }
        sc.close();
    }
    public static void main(String[] args) {
        try {
            readFile("input.txt");
        } catch (FileNotFoundException e) {
```

```
            System.out.println("Error: File not found!");
        }
    }
}
```

Q.26. Write a Java program to implement a lambda expression to find the sum of two integers.

```
Ans. interface AddOperation {
    int sum(int a, int b);
}
public class question26 {
    public static void main(String[] args) {
        AddOperation add = (a, b) -> a + b;
        int result = add.sum(10, 20);
        System.out.println("The sum is: " + result);
    }
}
```

Q.27. Write a Java program to create a basic Java thread that prints "Hello, World!" when executed.

```
Ans. class HelloThread extends Thread {
    @Override
    public void run() {
        System.out.println("Hello, World!");
    }
}
public class question27 {
    public static void main(String[] args) {
```

```
        HelloThread thread = new HelloThread();
        thread.start();
    }
}
```

Q.28. Write a Java program to create an array list, add some colors (strings) and print out the collection.

```
Ans. import java.util.ArrayList;
public class question28 {
    public static void main(String[] args) {
        ArrayList<String> a = new ArrayList<>();
        a.add("White");
        a.add("Black");
        a.add("Pink");
        a.add("Blue");
        a.add("Yellow");
        System.out.println(a);
    }
}
```

Q.29. Write a Java program to get the current date and time.

```
Ans. import java.time.LocalDateTime;
import java.time.format.DateTimeFormatter;
public class question29 {
    public static void main(String[] args) {
        LocalDateTime current = LocalDateTime.now();
        DateTimeFormatter formatter =
DateTimeFormatter.ofPattern("dd-MM-yyyy hh:mm:ss");
```

```
        String formatted = current.format(formatter);
        System.out.println("Current Date and Time:- " +
formatted);
    }
}
```

Q.30. Write a Java method to count all vowels in a string.
Test Data:
Input the string: Hello World!

Ans.
```
public class question30 {
    public static int countVowels(String str) {
        int count = 0;
        str = str.toLowerCase();
        for (int i = 0; i < str.length(); i++) {
            char ch = str.charAt(i);
            if (ch == 'a' || ch == 'e' || ch == 'i' || ch
== 'o' || ch == 'u') {
                count++;
            }
        }
        return count;
    }
    public static void main(String[] args) {
        String input = "Hello World!";
        int vowelCount = countVowels(input);
        System.out.println("Number of vowels in \"" +
input + "\": " + vowelCount);
    }
}
```